Saying less, steering more

Simulation 2 (asynchronous activation), setup `task003-nosolution`

LLM-free simulator; report of 14 October 2026; runs of 9 October 2026

Mirrors the Simulation 1 report on `task003-nosolution` of the same date, section by section.

How the simulator works. The world, `task_003`, has three candidate allocations; A0 is the correct one. Each of 24 agents is given one true fact about it and reasons as an exact Bayesian: its *belief* is the exact probability of A0, A1 and A2 given the facts it currently remembers. There are no rounds. Each agent acts at random times, on average once per time unit: it reads q posts drawn at random from the 48 most recent posts on a shared *board* (never its own), adds their facts to its memory, recomputes its belief, *votes* by probability matching (a vote drawn from its belief) and posts one remembered fact that supports its vote, or nothing if none does. Every remembered fact is forgotten after a random lifetime (on average a fraction ρ survives one time unit; reading it again renews it). A *controller* with a target, A0 (truth) or A2 (false), acts at random times at rate λ_c until $t = 30$: it reads q_c recent agent posts and, unless the *vote gate* (at least 75% of them from agents voting for its target) or the *proof stop* (its target already proved) silences it, posts the one fact from its pool that best supports its target, at most 30 messages per episode. Agents continue alone until $t = 40$. Every episode is run three times with the same random draws: without a controller, with the truth controller and with the false controller.

Summary. 24 agents hold evidence that, pooled, is a coin flip between allocation A0 (the truth) and A2, with no proof available to anyone. Simulation 2 removes the day/night rounds of Simulation 1: agents and the controller act at random times, remembered facts fade continuously, and the shared board fades too. Both controllers still tip the swarm, and **the effect persists** after they stop. Measured in votes, control is **about as strong as in Simulation 1**. The controller's rate barely matters when agents forget. With full memory, **a faster controller steers less**: it puts more of its pool into circulation, and the whole pool pulls every belief toward the same point whatever the target. Turning the vote gate off adds about 16 messages per episode and no control. **Correction to the 9 and 12 October reports**: measured in mean belief, control looked weaker than in Simulation 1 when agents forget (0.29 against 0.42); measured in *votes*, as in Simulation 1, the gap almost disappears (0.40 against 0.42; false controller 0.25 against 0.29). Most of the difference came from when beliefs are measured (Section 2). Measured by whether the swarm's *collective decision* is the target, **the cheapest controller already reaches the full gain** (about 7 facts posted), and most settings spend two to four times more for nothing; only early on ($t = 5$) does speed buy more. Main hypotheses were confirmed on a second, independent set of random draws.

1 Setup

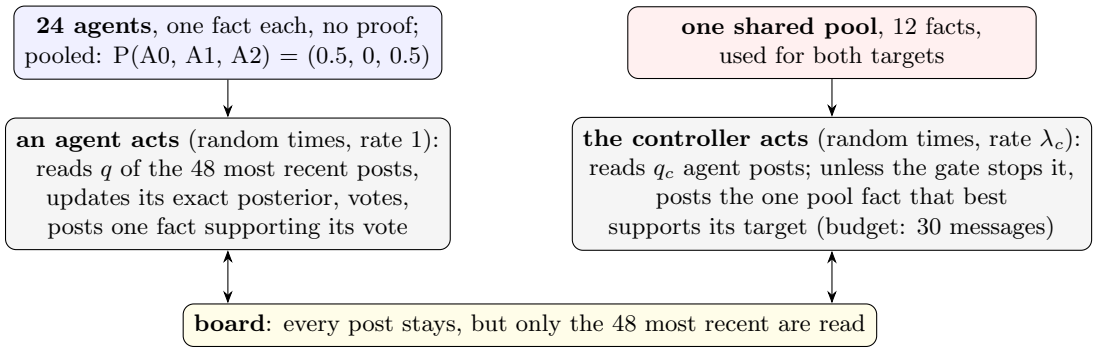


Diagram 1. Simulation 2. There are no rounds. Each remembered fact is forgotten after a random lifetime (on average a fraction ρ survives one time unit) and comes back if read again. The controller acts for $0 \le t < 30$; agents continue alone until $t = 40$.

The agents' facts plus the *whole* pool give P(A0, A1, A2) = (0.75, 0, 0.25): the truth keeps an edge that cannot be designed away, because every fact is true. Agents are exact Bayesian reasoners and vote by probability matching (the vote is drawn from the posterior). The *vote gate* keeps the controller silent when at least 75% of the posts it read come from agents who voted for its target. The grid: posts read $q \in \{3, 6, 12\}$, $q_c \in \{3, 6, 12, 24\}$, memory $\rho \in \{0.75, 1\}$, controller rate $\lambda_c \in \{0.5, 1, 2, 4, 8\}$ actions per time unit, gate on/off; 1,000 episodes per

setting. Control is measured by the *gain*

$$G(t) = \frac{1}{E}\sum_{e=1}^{E}\Big[m_{\text{target}}(t)^{\text{ctrl}}(e) - m_{\text{target}}(t)^{\text{silent}}(e)\Big], \tag{1}$$

where $m_k(t)$ is the agents' mean belief in allocation k and each episode e uses the same random draws with and without the controller. (Simulation 1 measures the gain in the share of *votes*; Figure 1 shows the vote shares of both simulations on the same scale.) Unless stated otherwise: $q = 6$, $q_c = 12$, $\lambda_c = 1$, gate on.

2 Main result: both controllers tip the swarm, and the effect stays

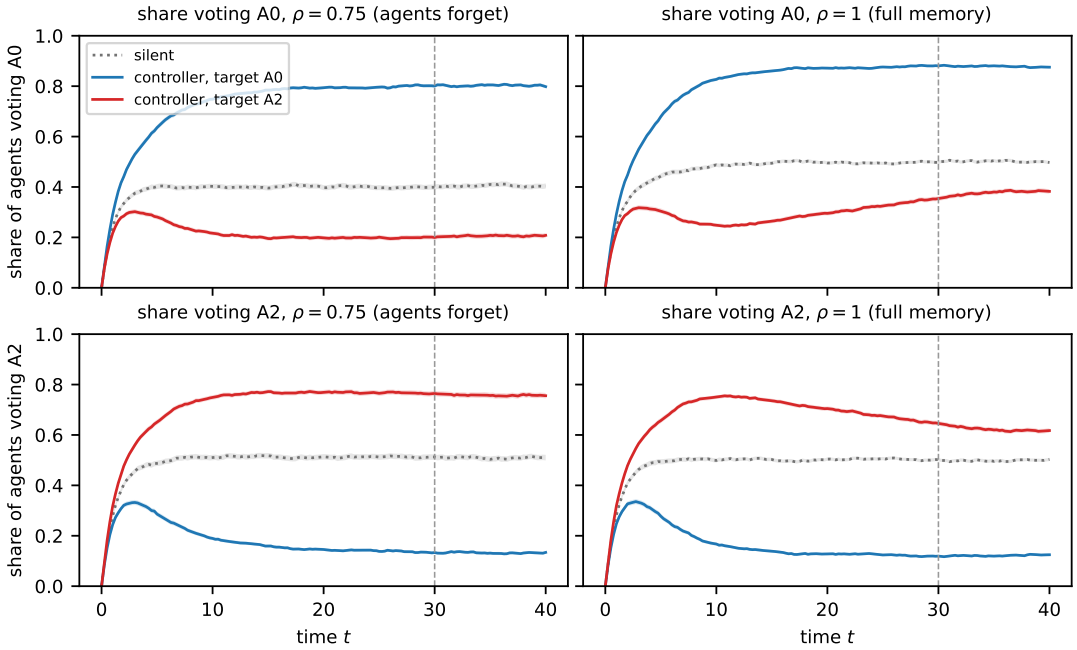


Figure 1: Share of the 24 agents voting A0 (top) and A2 (bottom) over time, without a controller (dotted grey), with the controller targeting A0 (blue) and with the one targeting A2 (red); $\rho = 0.75$ (left) and $\rho = 1$ (right). Agents who have not acted yet have no vote, so the shares start at 0 (all agents have voted by $t \approx 5$). The controller stops at $t = 30$ (dashed). Bands: 95% intervals. $q = 6$, $q_c = 12$, $\lambda_c = 1$, gate on.

- **Without a controller the swarm stays split**: at $t = 30$, 40% vote A0 and 51% A2 when agents forget, 50% each with full memory.
- **Either controller tips it within about ten time units**: the target's share reaches 0.80 (A0) or 0.77 (A2) when agents forget, and 0.88 (A0) or 0.65 (A2) with full memory.
- **With full memory the false controller's lead erodes**: the A2 share peaks at 0.76 ($t \approx 11$) and falls to 0.62 by $t = 40$, as agents accumulate the evidence that favours A0.

Table 1: Gain at $t = 30$, $\lambda_c = 1$, gate on: in mean belief, Eq. (1) (in brackets the silent swarm's belief in the target), and in the share of votes for the target, the measure of Simulation 1. Simulation 1: one fact per night, same q, q_c, ρ (there the two measures agree within 0.01).

		Simulation 2		Simulation 1	
ρ	controller	belief	votes	votes	messages sent
0.75	truth (A0)	0.29 (0.38)	0.40	0.42	11
0.75	false (A2)	0.19 (0.48)	0.25	0.29	12
1	truth (A0)	0.38 (0.50)	0.38	0.38	7
1	false (A2)	0.14 (0.50)	0.14	0.15	15

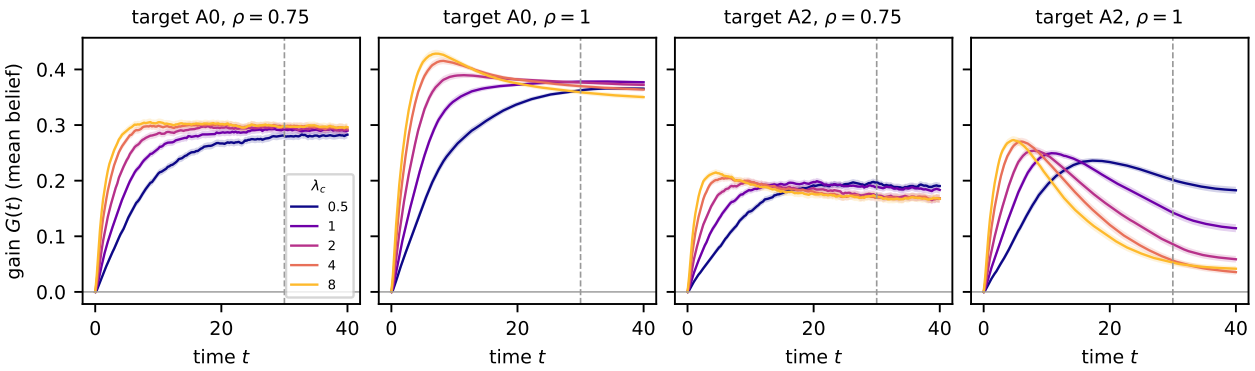


Figure 2: Gain over time, Eq. (1), one line per controller rate λ_c; gate on. The controller stops at $t = 30$ (dashed). Bands: 95% intervals.

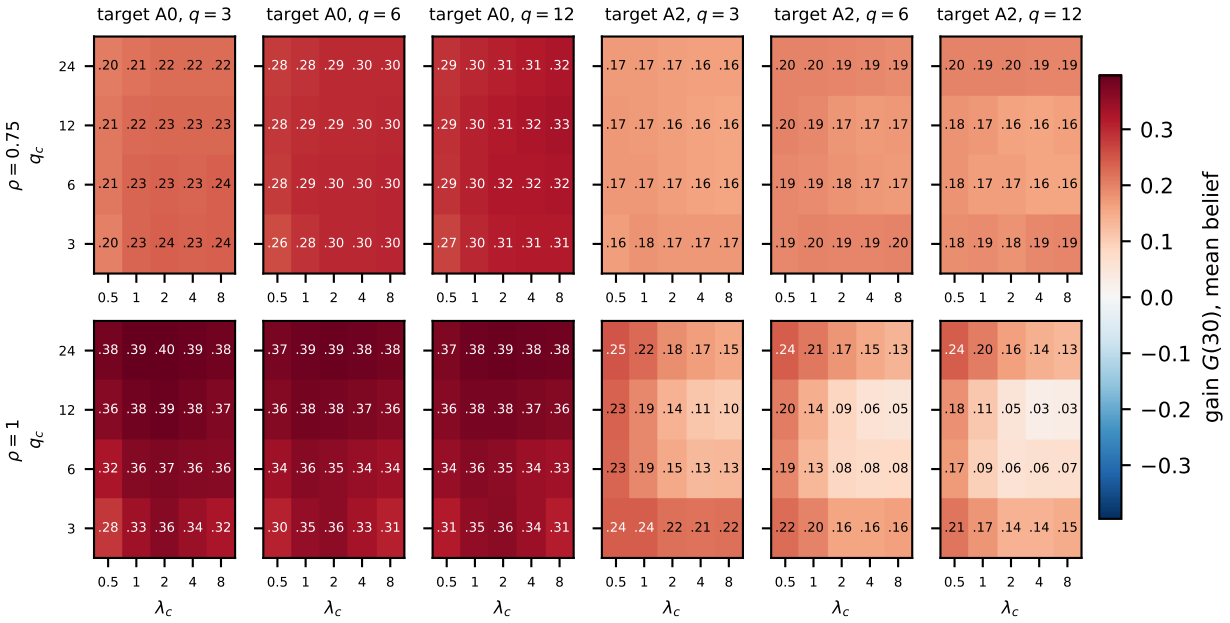


Figure 3: Gain $G(30)$ over the controller rate λ_c (x) and the posts it reads q_c (y), for each target and every q (columns), $\rho = 0.75$ and 1 (rows); gate on. Grey number: the 95% interval includes 0.

- **Both controllers tip the swarm** (Table 1); the truth controller's gain is larger, mostly because the silent swarm already leans to A2 when agents forget (Figure 1): there is more room to raise A0.
- **In votes, control is about as strong as in Simulation 1**: 0.40 against 0.42 (truth) and 0.25 against 0.29 (false) when agents forget; with full memory the two agree within 0.01. **In mean belief it looks weaker** when agents forget, by 0.13 (truth) and 0.09 (false), a difference confirmed with fresh random draws. The two measures part because an agent votes right after reading, with a fresh memory, and then keeps forgetting until it acts again, so a belief snapshot at a fixed time catches agents with faded memories (truth controller, $\rho = 0.75$: 80% vote A0 at $t = 30$, but the mean belief in A0 is 0.67; in Simulation 1 the day-end snapshot comes before dawn's forgetting, and beliefs and votes agree). Earlier reports read the belief gap as weaker control; the vote gap says the collective decision is affected much less.
- **The effect persists**: ten time units after the controller stops, the gain has changed by at most 0.016 at $\rho = 0.75$ (Figure 2); agents keep relaying the controller's facts (12–45% of their posts). With full memory a fast false controller peaks early and then loses most of its gain (Section 3).

3 The controller's strength and the vote gate

3.1 Strength: the controller's rate

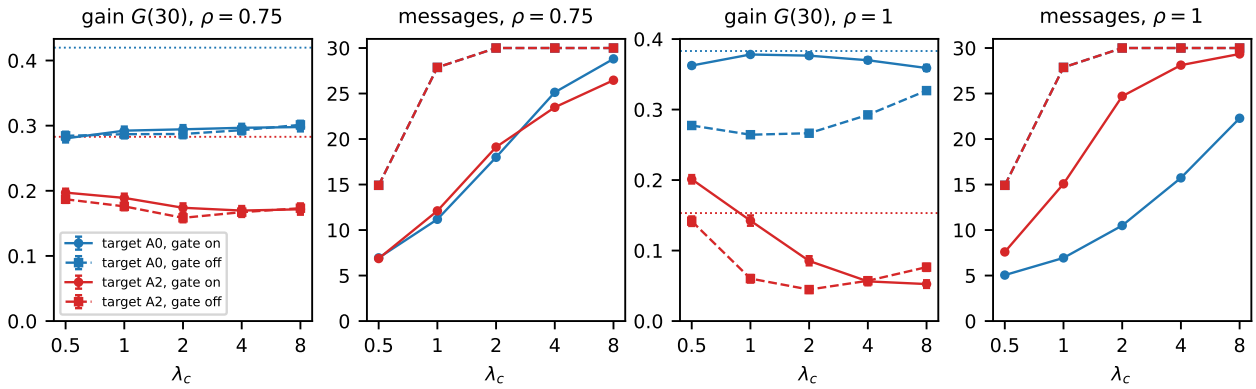


Figure 4: Gain at $t = 30$ and messages sent against the controller rate, for $\rho = 0.75$ and $\rho = 1$ (target A0 blue, A2 red; gate on solid, off dashed; dotted: Simulation 1).

When agents forget, the rate barely matters. From $\lambda_c = 0.5$ to 8 the gain moves from 0.28 to 0.30 (truth) and from 0.20 to 0.17 (false), but the messages sent rise from about 7 to 26–29. A fast controller spends its budget early (by $t \approx 4$ with the gate off), reaches its gain sooner, and then the gain stays put. Gain per message is highest for a slow, gated controller.

With full memory, a faster controller steers less. The false controller's gain falls from 0.20 at $\lambda_c = 0.5$ to 0.05 at $\lambda_c = 8$, and with the gate off the truth controller loses 0.05 on average against the gate on, up to 0.12 (Figure 4, right). The same fall appears at every q and at $q_c = 3$, 6 and 12 (Figure 3, bottom row).

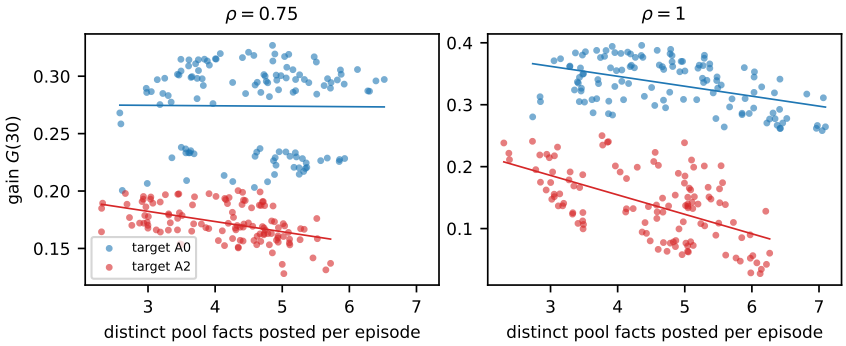


Figure 5: Each point is one setting (q, q_c, λ_c, gate): gain at $t = 30$ against the number of different pool facts the controller posted per episode. Lines: least-squares fits.

Why: saying more of the pool pulls toward the same point. Every fact is true, and the agents' facts plus the whole pool give (0.75, 0, 0.25). With full memory nothing posted is forgotten, so the more *different* pool facts a controller puts into circulation, the closer beliefs move to that point, whatever its target: down from the false controller's side, and down from the truth controller's best, which lies above 0.75. At $\rho = 1$ each additional distinct fact lowers the gain by 0.031 (false) and 0.016 (truth) (Figure 5, right). The rate sets that number: spread over 30 time units, the controller reads varied boards and posts more different facts (6.3 per episode at $\lambda_c = 1$, gate off); at $\lambda_c = 8$ it spends its budget in a burst on similar boards and repeats itself (5.0), which is why the gain partly recovers there. When agents forget, facts leave again and the effect is weak (Figure 5, left). This was found after looking at the data, so it was rerun with fresh random draws: **every one of the 16 checked reversals reappeared** (false controller, gain at $\lambda_c = 0.5 \ldots 8$: 0.20, 0.15, 0.08, 0.05, 0.05). The explanation by distinct facts rests on a correlation across settings and is not yet tested directly (for example with a controller that remembers what it has said). Simulation 1 has a different route to the same outcome: there a large posting budget forces facts against the target onto the board.

3.2 The vote gate

The gate keeps the controller silent when it reads enough support. Turning it off adds about 16 messages per episode at $\lambda_c = 1$ (confirmed with fresh draws) and **no control** when agents forget (-0.017 and -0.003); with full memory it *lowers* the gain (truth -0.05 on average) through the mechanism above.

4 The controller overestimates its support

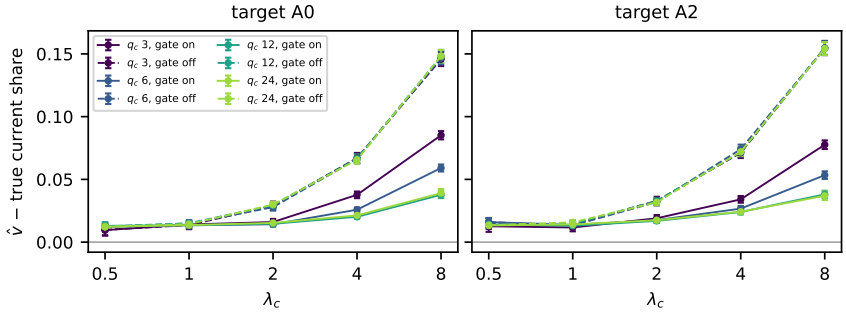


Figure 6: Error of the controller's estimate $\hat{v}$ (share of posts read whose author voted for the target) against the true current share of all 24 agents' last votes; $\rho = 0.75$; solid gate on, dashed off.

Unlike in Simulation 1, $\hat{v}$ is biased upward, and **the faster the controller, the larger the bias**: about $+0.014$ at $\lambda_c \leq 1$, rising to $+0.05$ (gate on) and $+0.15$ (gate off) at $\lambda_c = 8$. A likely reason, not yet tested: the posts it reads come from agents who acted recently, often just after reading the controller's facts, while the true share also counts older votes. More posts read (q_c) reduce the error's spread (0.16 at $q_c = 3$, 0.07 at $q_c = 24$) but not the bias.

5 How readable is the command? Information and pair efficiency

A controller can change the swarm in two ways: in the direction it is asked to, or in ways that happen whatever its target. We measure how readable the command is in the swarm (the mutual information between the target and the swarm's state), which share of the change is directed (the pair efficiency η), and how unsure the agents become. Throughout, the target is chosen by a fair coin, and all quantities are taken at one time t, across the 1,000 episodes of each arm (not over whole trajectories). Setting: $q = 6$.

5.1 Measures

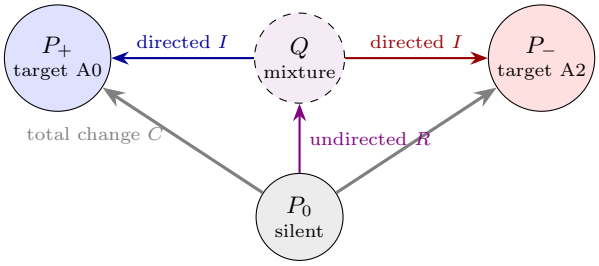


Diagram 2. Three distributions of the swarm's state at time t, one per arm, each over the episodes of that arm. The controller's total change C splits exactly into a part that depends on its target (I, from the mixture Q to each arm) and a part that does not (R, from silence to Q). Efficiency $\eta = I/C$ is the directed share.

Distributions. For each arm a (silent 0, target A0 $+$, target A2 $-$) and a time t, each episode e contributes a probability vector $y_a(e)$ over the values x of the state X: at *one-agent* level its vote shares (the vote of a randomly chosen agent), at *swarm* level a mark on the bin of $s = m_{\mathrm{A0}} - m_{\mathrm{A2}}$ (difference of mean beliefs, three equal bins on $[-1, 1]$). Then $\hat{P}_a = \frac{1}{n}\sum_e y_a(e)$. Any pair of controllers can be written as

$$\hat{P}_\pm = \hat{P}_0 + \mu \pm \delta, \qquad \mu = \tfrac{1}{2}(\hat{P}_+ + \hat{P}_-) - \hat{P}_0, \quad \delta = \tfrac{1}{2}(\hat{P}_+ - \hat{P}_-), \tag{2}$$

a *common* part μ (what control does on average, whatever the target) and a *directional* part δ; both are vectors over the values of X (all answers, A1 included).

Readability and the pair efficiency. With $\hat{Q} = \hat{P}_0 + \mu$ and $\mathrm{KL}(p\|q) = \sum_x p(x)\log_2[p(x)/q(x)]$,

$$I = \tfrac{1}{2}\mathrm{KL}(\hat{P}_+\|\hat{Q}) + \tfrac{1}{2}\mathrm{KL}(\hat{P}_-\|\hat{Q}), \quad C = \tfrac{1}{2}\mathrm{KL}(\hat{P}_+\|\hat{P}_0) + \tfrac{1}{2}\mathrm{KL}(\hat{P}_-\|\hat{P}_0) = I + R, \quad R = \mathrm{KL}(\hat{Q}\|\hat{P}_0), \quad \eta = I/C. \tag{3}$$

I is the mutual information between the target and the state: how readable the command is (at most 1 bit). With the one-agent state it is $I(\text{target}; \text{vote of a randomly chosen agent})$, the quantity mapped in Figure 9. $I = \frac{1}{2}\mathrm{KL}(\hat{Q}+\delta\|\hat{Q}) + \frac{1}{2}\mathrm{KL}(\hat{Q}-\delta\|\hat{Q})$ depends on δ and on where the pair sits ($\hat{Q}$); $R = \mathrm{KL}(\hat{P}_0+\mu\|\hat{P}_0)$ depends only on μ; C (a number) $= I + R$, so $R = C - I$. The reference $\hat{P}_0$ is the silent arm. For small changes, $\mathrm{KL}(p+\epsilon\|p) \approx \sum_x \epsilon(x)^2/(2\ln 2\, p(x))$ gives

$$I \approx \frac{\|\delta\|^2}{2\ln 2}, \quad R \approx \frac{\|\mu\|^2}{2\ln 2}, \quad \eta \approx \frac{1}{1 + \|\mu\|^2/\|\delta\|^2}, \qquad \|v\|^2 = \sum_x \frac{v(x)^2}{\hat{P}_0(x)}. \tag{4}$$

So $\eta = 1$ exactly when the two controllers' changes are mirror images ($\mu = 0$), whatever their size, and η falls only through the *ratio* of common to directional change: when both push the same way (a common push), or one moves the swarm much more than the other (a lopsided pair: along one direction with sizes a and b, $\|\mu\|/\|\delta\| = |a-b|/(a+b)$). It is a property of the pair, not of either controller, and it stays constant whenever both effects grow together; the size of the effect is measured by I. At swarm level C is often infinite (the controlled swarm reaches states silence never visits); there we use $\eta_{JS} = I/(I + \mathrm{JS}(\hat{Q}, \hat{P}_0))$ with the bounded Jensen–Shannon divergence; $\mathrm{JS}(\hat{Q}, \hat{P}_0)$ is the information the state carries about *whether* a controller is present (controlled or silent, 50/50), so η_{JS} compares information about the direction with information about the presence of control.

The agents' own uncertainty: $\Delta\bar{H}$ (change in the mean entropy of their beliefs) and ΔJSD (change in disagreement, the entropy of the mean belief minus $\bar{H}$), controlled minus silent, same episode.

5.2 Over time

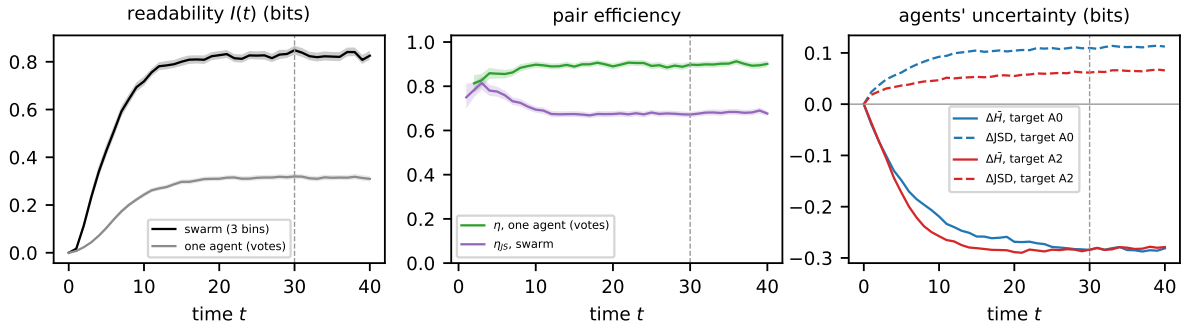


Figure 7: At $\rho = 0.75$, $q_c = 12$, $\lambda_c = 1$, gate on (the controller stops at $t = 30$, dashed): readability $I(t)$ at swarm and one-agent level, pair efficiency (η, one agent; η_{JS}, swarm), and the change in the agents' uncertainty. Bands: 95% intervals over episodes.

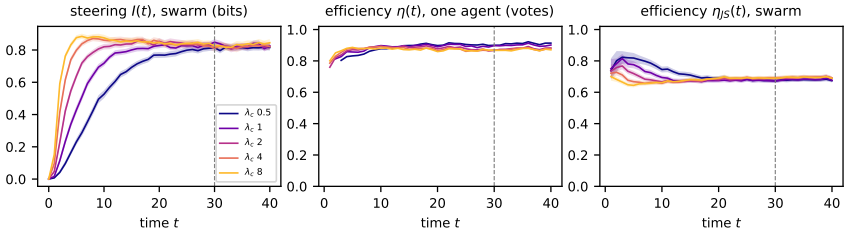


Figure 8: Readability and pair efficiency over time, $\rho = 0.75$, $q_c = 12$, gate on, one line per rate; the controller stops at $t = 30$ (dashed). Bands: 95% intervals.

- **Readability builds within 10–15 time units and persists**: at $t = 30$, $I = 0.85$ bits (swarm) and 0.32 bits (one agent's vote), almost unchanged at $t = 40$; $\eta = 0.90$, $\eta_{JS} = 0.67$. Faster controllers get there sooner; all rates end at the same level.
- **Agents become surer** ($\Delta\bar{H} = -0.28$ bits) **but disagree more** (ΔJSD +0.11 for target A0, +0.06 for target A2).

5.3 Across the control plane

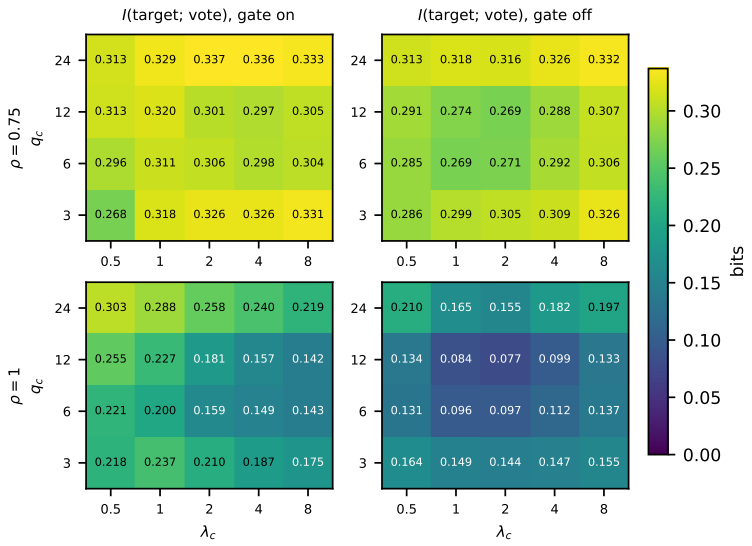


Figure 9: Readability $I(\text{target};\text{vote})$ at $t = 30$, in bits, over the reading budget q_c (y) and the controller rate λ_c (x); state: the vote of a randomly chosen agent.

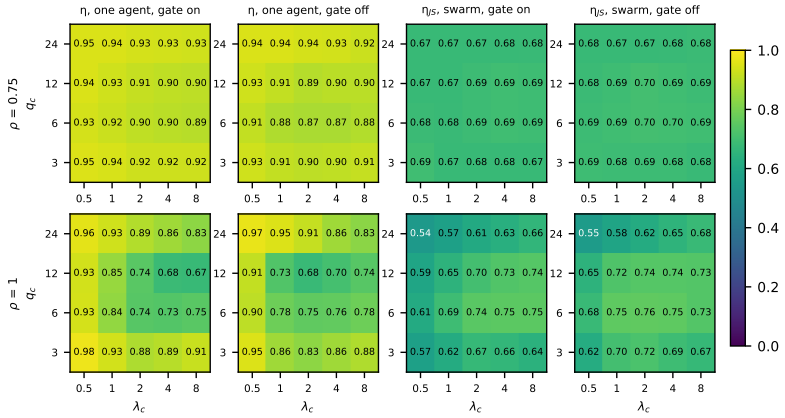


Figure 10: Pair efficiency at $t = 30$ over q_c and the controller rate λ_c: one-agent η (beliefs) and swarm η_{JS}. Dash: undefined.

- **When agents forget, readability hardly depends on the settings**: 0.27–0.34 bits for every q_c, rate and gate (Figure 9, top), and η is flat (0.86–0.95).
- **With full memory readability falls with the rate**: from 0.30 bits ($q_c = 24$, $\lambda_c = 0.5$) to 0.14–0.22 at $\lambda_c = 8$ with the gate on, and lower still with the gate off (0.08–0.21). This is the false controller's stall (Section 3): its target becomes harder to tell from the truth controller's.

5.4 Readability, pair efficiency and uncertainty against the controller rate

- **When agents forget, η is flat (0.86–0.95)**: the two controllers' effects grow together.
- **With full memory η falls with the rate** (0.93 to 0.67) because the pair turns lopsided: the false controller's gain collapses, the truth controller's does not. Swarm η_{JS} moves the other way, because swarm readability is saturated at 1 bit; the one-agent measures are the informative ones here.

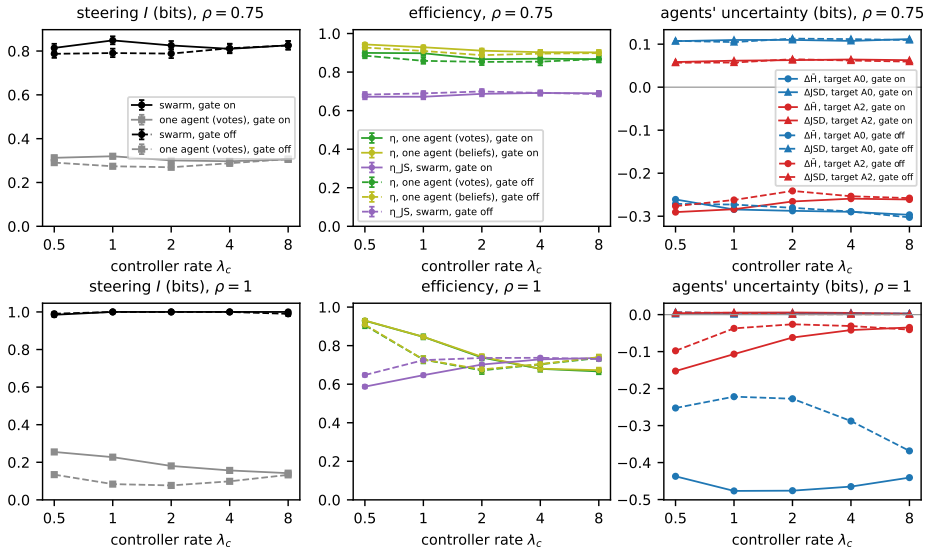


Figure 11: Readability I (left), pair efficiency (middle) and the change in the agents' uncertainty (right; circles $\Delta\bar{H}$, triangles ΔJSD) at $t = 30$ against the controller rate λ_c ($q_c = 12$; solid: gate on; dashed: gate off). Bars: 95% intervals.

5.5 Precision at 50, 100 and 300 paths

Table 2: Value from 1,000 paths per arm, and root-mean-square error of the estimate from 50, 100 or 300 paths (200 random subsets each), at $t = 30$, $\lambda_c = 1$, $q_c = 12$, gate on, $\rho = 0.75$.

quantity	1,000 paths	error, 50	error, 100	error, 300
steering I, swarm (3 bins)	0.849	0.039	0.026	0.013
steering I, one agent (votes)	0.320	0.023	0.017	0.009
efficiency η, one agent (votes)	0.898	0.032	0.022	0.010
efficiency η_{JS}, swarm	0.672	0.027	0.019	0.010
$\Delta\bar{H}$, target A0	-0.284	0.024	0.016	0.009
$\Delta\bar{H}$, target A2	-0.284	0.023	0.016	0.008
ΔJSD, target A0	0.110	0.008	0.006	0.003
ΔJSD, target A2	0.062	0.008	0.006	0.003

Across all settings, at 100 paths: readability ± 0.01 (one agent) to ± 0.03 (swarm), η ± 0.02–0.03, $\Delta\bar{H}$ ± 0.02, ΔJSD ± 0.006. Two settings that differ by at least 0.02 are ordered correctly in 90–100% of subsets.

6 Success against facts posted

The information measures above say how readable the command is, not whether the swarm ends up deciding what the controller wants, nor at what cost. Here we measure both directly.

- **Collective decision** Y_T: plurality of the agents' last votes at time T; a tie between k answers counts $1/k$. **Success**: $Y_T =$ the controller's target.
- **Benefit** $\Delta_\pi = p_\pi - p_0$: change in the probability that the target wins, against the silent arm of the same episodes; p_0 is the silent probability.
- **Cost** C_π: facts actually posted up to T, mean per run (not the allowed budget).
- **Best-achievable curve** among the tested settings π of one group (ρ, target): $\Delta^*(B) = \max\{\Delta_\pi : C_\pi \leq B\}$, the largest gain reached spending at most B; read the other way, $C^*(\delta) = \min\{C_\pi : \Delta_\pi \geq \delta\}$, the least spending that reaches δ. A setting below the curve is *dominated*: another reached at least as much for less.
- **Cost efficiency** $\eta_{\text{cost}} = C^*(\Delta_\pi)/C_\pi$: the share of the spending that was necessary (1: nothing cheaper reached as much). Not defined where $\Delta_\pi \leq 0$ or its 95% interval includes 0.

How to read the curves. Each dot is one setting, at (facts posted, gain). The **solid line** is the best-achievable curve over the tested settings: it starts at $(0, 0)$ (no control) and steps up each time a setting reaches a higher gain than anything cheaper; dots on it have $\eta_{\text{cost}} = 1$, dots below it are dominated, and η_{cost} is how far to the left the curve reaches the same gain. The **dashed line** is what becomes reachable if the controller *chooses its setting at random for each run*: using setting X (7 facts, +0.32) in half of the runs and no control in the other half costs 3.5 facts and gains +0.16 *on average*, so every point on the straight segment between two settings (or between a setting and $(0, 0)$) is reachable on average; the dashed line is the outermost such segments (the concave hull through $(0, 0)$). It lies above the solid line where mixing beats every single setting. Its first segment from $(0, 0)$ is arithmetic, not an observation.

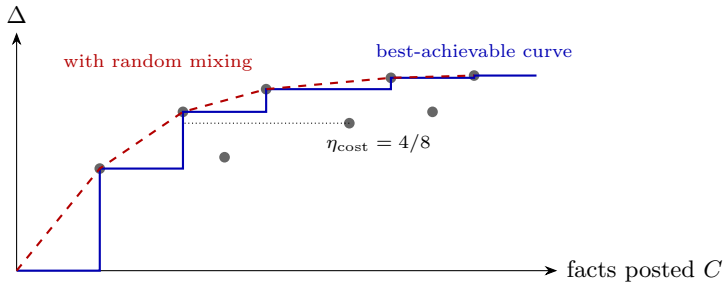


Diagram 3. Illustration. Each point is a setting. Solid: best-achievable curve; dashed: reachable on average by random mixing per run. The setting at cost 8 reaches $\Delta = 0.65$, which a setting costing 4 already reaches: $\eta_{\text{cost}} = 0.5$.

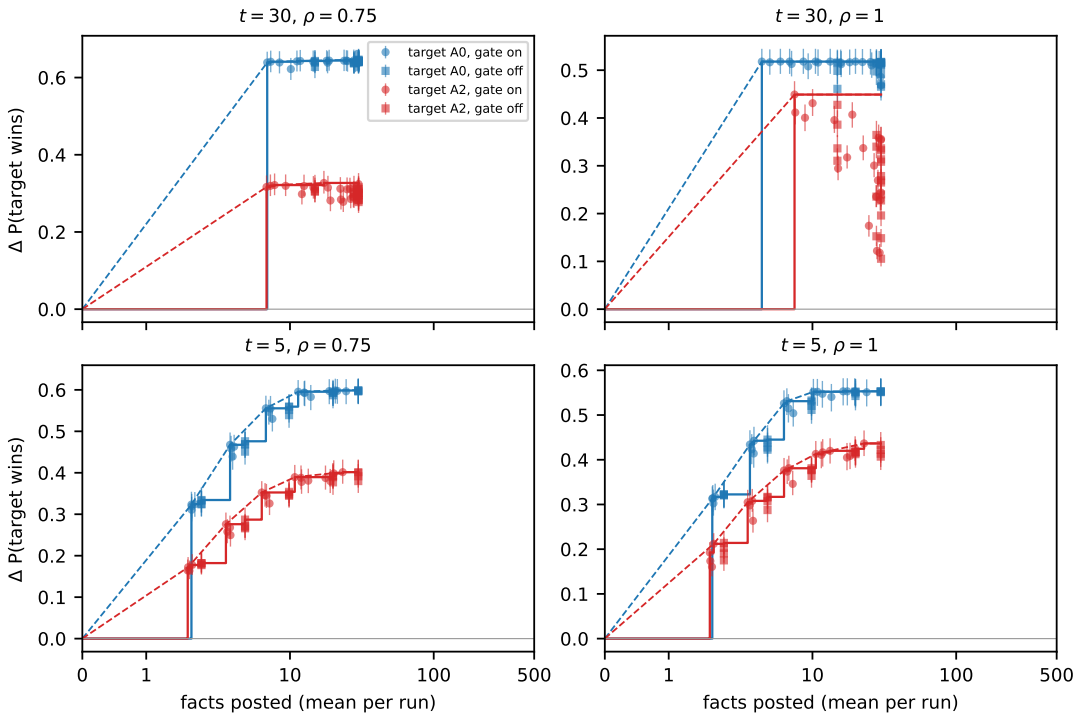


Figure 12: Change in the probability that the controller's target wins against facts posted, at $t = 30$ (top) and $t = 5$ (bottom), $\rho = 0.75$ (left) and 1 (right); every setting ($q_c \times \lambda_c \times$ gate, $q = 6$) with its 95% interval; blue: target A0, red: target A2; circles gate on, squares gate off. Solid: best-achievable curve; dashed: with random mixing per run. The x axis is linear from 0 to 1 and logarithmic above, up to 500 as in Simulation 1.

- **At $t = 30$ success saturates with very little control.** When agents forget, every setting reaches about the same gain (+0.62 to +0.65 for target A0, +0.28 to +0.33 for target A2), whatever it spent (7 to 30 facts). The cheapest setting ($\lambda_c = 0.5$, gate on, $q_c = 12$; 6.9 facts) is already on the flat part of the curve; the median setting spends two to four times more for the same result (median η_{cost} 0.48 for target A0, 0.26 for A2; lowest 0.23, gate off at high rates). Spending after the decision is settled buys nothing.
- **With full memory only one setting is efficient** for each target ($q_c = 24$, $\lambda_c = 0.5$, gate on: 4.4 facts for +0.52, target A0; 7.5 facts for +0.45, target A2). For target A2 more spending is worse: the gain falls to +0.11 at 30 facts (the rate reversal of Section 3).

- **Early on, speed buys success, with diminishing returns.** At $t = 5$ the gain rises with the facts posted: target A2 from +0.16 (1.9 facts) to +0.40 (23 facts), target A0 from +0.32 to +0.60; the first few facts buy most of it, and 14–16 of the 40 settings lie on the curve. By $t = 10$ the slow controllers have caught up, so efficiency depends on when success is scored.
- **Target A0 gains more than A2 mostly because of where the silent swarm starts**: without control A2 wins 64% of runs and A0 35% when agents forget (Figure 1). As a share of the room left, $\Delta/(1 - p_0)$, the two targets are closer (median 0.99 for A0, 0.84 for A2).
- **Precision at 100 paths**: Δ ±0.017; η_{cost} ±0.02–0.035. Where gains are flat, which setting is "on the curve" is decided by differences smaller than this, so single values of η_{cost} near 1 among equal gains should not be over-read.

Caveats. One world (task_003), 24 agents, exact-Bayesian agents, probability-matching votes; a controller without memory that posts one fact at a time; a budget of 30 messages. Means over 1,000 episodes per setting. Efficiencies are relative to the tested settings: nothing was tested below about 2 facts ($t = 5$) or 7 facts ($t = 30$). Full analysis, code and tables: `analysis/task003_llm_free/simulation_2/analysis/` (report of 9 October, reviewer guide) and `analysis/task003_llm_free/efficiency_estimates/`.